

Perfect hindsight would solve up to 16 more tasks in 100 than the best single view. None of 8 learned choices beat always using the cheapest view on both success and cost.

RESULTS ACROSS 8,934 TASK ATTEMPTS

+16.35 in 100

PERFECT HINDSIGHT, BEST OF 8 · VS ONE FIXED VIEW

0 of 8

LEARNED CHOICES THAT BEAT ALWAYS-CHEAPEST

1 of 8

HINDSIGHT CHOICES THAT BEAT ALWAYS-CHEAPEST

ONE FIXED VIEW — the single view that solves most tasks in a setting, used for every task.

ALWAYS-CHEAPEST — the single view that costs least on average in a setting, used for every task.

PERFECT HINDSIGHT — for each task, the view that solved it, picked after the fact. An optimistic bound: rerunning one unchanged view flips 10–14% of outcomes and by itself gains 2.0–7.6 tasks in 100.

THE CATCH

The agents that would gain most from choosing a view solve the fewest tasks — so they produce the fewest examples to learn the choice from.

WHY LEARNING THE CHOICE FAILS HERE

A training example for “which view” exists only when the agent solves a task. Here the best single view solves just **2–36%** of tasks, leaving typically **15–97** usable examples per setting — enough to train a classifier in only 2 of the 6 VisualWebArena settings.

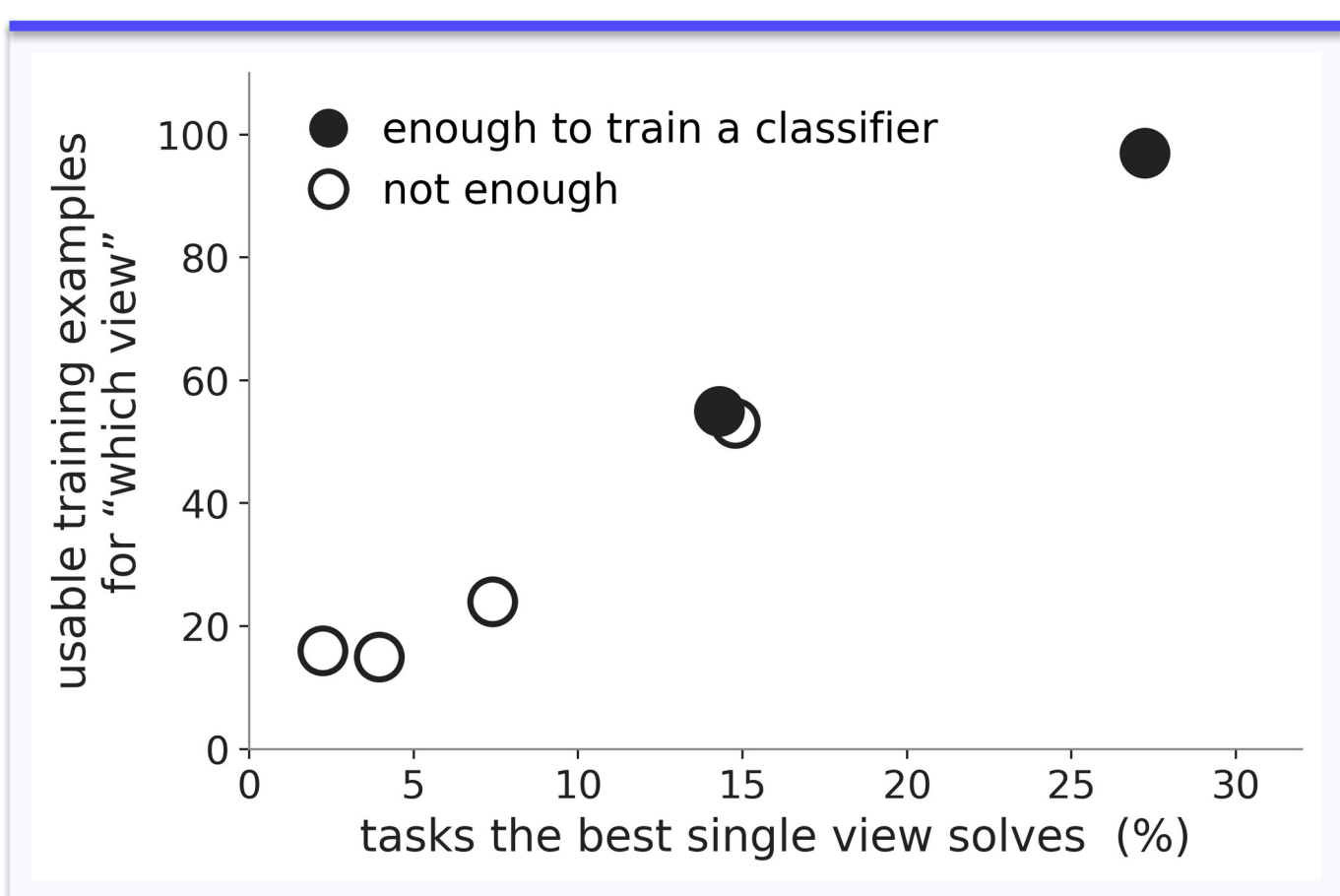


Fig 3. Usable “which view” examples against the best single view's success rate, one point per VisualWebArena setting. Examples exist only where tasks get solved.

Shrinking the training data on purpose points to scarcity as the main bottleneck, and prices it: the failing settings would need at least **2.1–4.2×** more tasks than the benchmarks contain.

TAKEAWAY

Learning how to see is not only a modelling problem: whether it can be learned depends on how good the agent producing the examples already is. So, in this order: improve the agent, then collect reliable examples, then learn when to look.

Measured in the 2–36% success regime we observed. This need not hold for stronger agents.

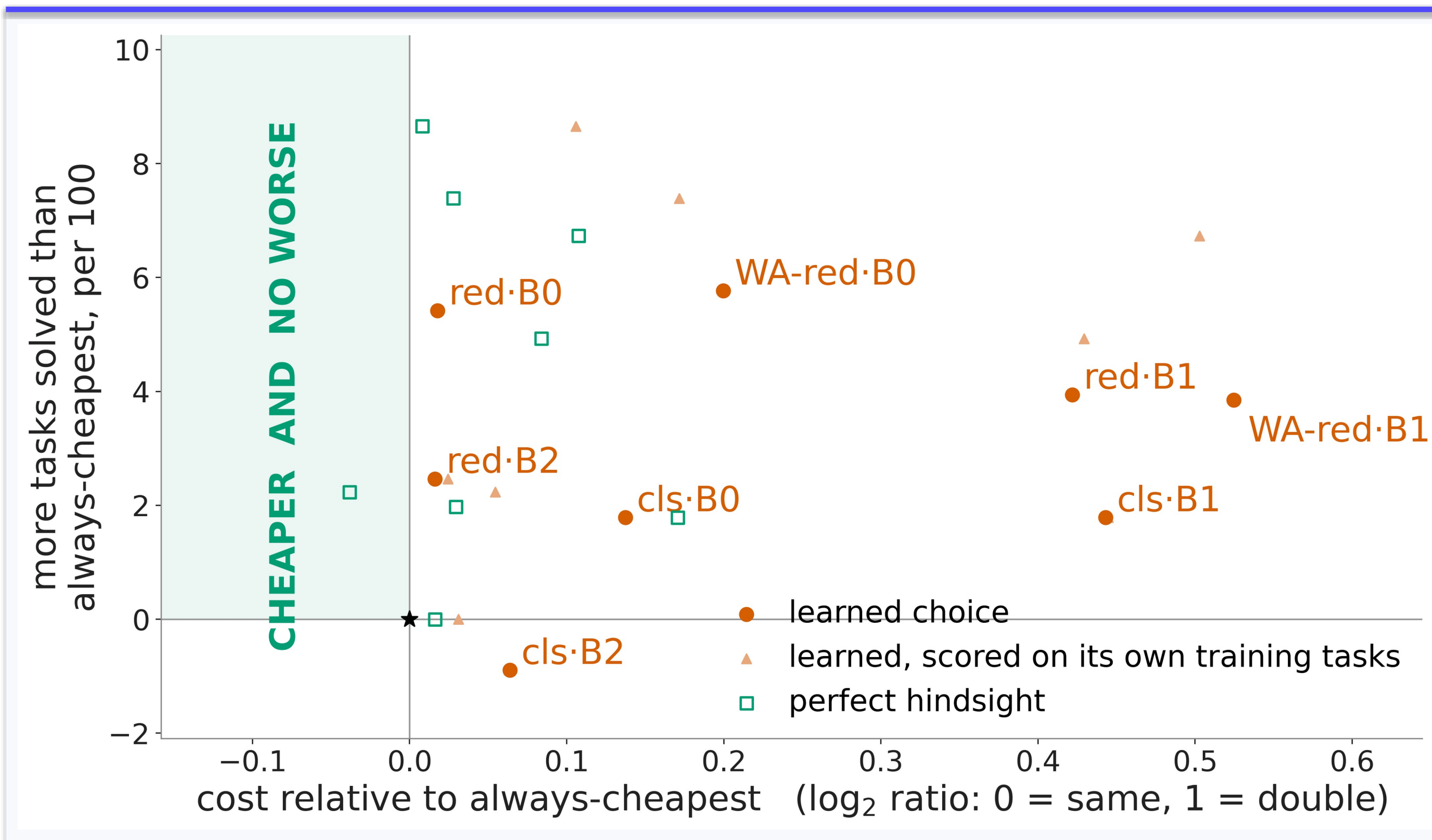


Fig 2. Every way of choosing a view, in every setting, compared with one fixed rule: **always use the cheapest view** (★). A win lands in the shaded region — cheaper *and* no worse. Always-cheapest is cheapest on average, not on every task, which is why a few points sit left of it. Learned choices are scored only on tasks they never saw.

Perfect hindsight would pay. Picking the winning view for each task after the fact would solve **3.45 to 16.35 more tasks in every 100** than the best single view, and spend 1.6–35.3% less — in all 8 settings.

Nothing we trained could do it. In **0 of 8** settings did a learned choice beat always using the cheapest view on both success and cost — and even perfect hindsight manages that in only **1 of 8**.

What survives is a bound, not a method. Sending the tasks nobody solves to the cheapest view saves 9.5–30.6% at the same success in 8 of 8 — but that too needs hindsight, and plain always-cheapest usually saves more.

WATCH THE LAPTOP BESIDE THIS POSTER

Same task, three views, different behaviour and different bills — step by step. Three illustrative tasks, chosen so each view wins once; **not** how often each wins.

	LOOK	READ	BOTH
“Navigate to the item on this page whose image is taken during a sunset.”	✓ 2 steps · \$0.007	✗ 21 steps · \$0.096	✓ 3 steps · \$0.014
“Navigate to my listing of the blue bike and change the price to \$85.50 (including in the description).”	✗ 30 steps · \$0.116	✓ 9 steps · \$0.079	✗ 6 steps · \$0.029
“Show me the cheapest bike with red handlebars between \$900-950.”	✗ 19 steps · \$0.075	✗ 6 steps · \$0.025	✓ 7 steps · \$0.037

One recorded attempt per view, B0 · classifieds. Every ✓ / ✗ came out the same on an independent rerun; steps and bill differ run to run.

HOW WE MEASURED IT — THE AGENT LOOP

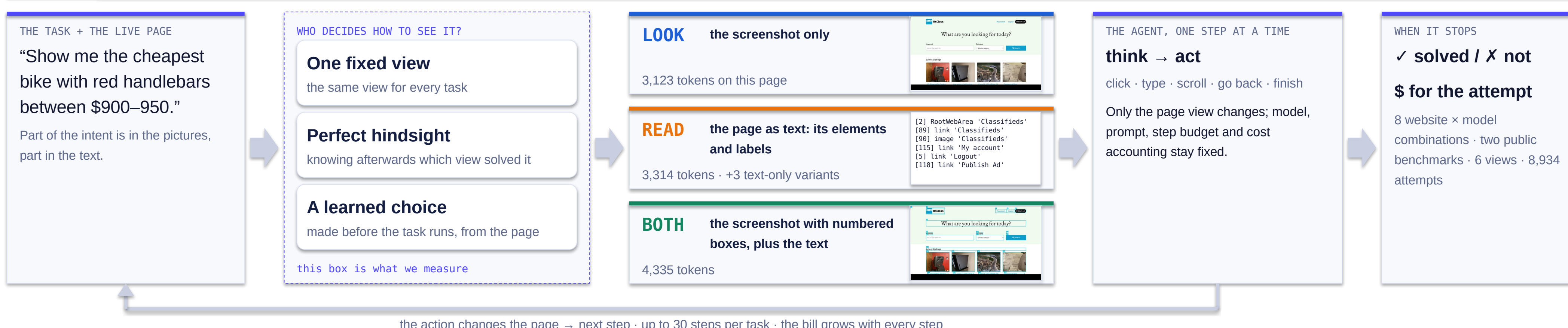


Fig 1. Everything is held fixed except how the page is shown. The dashed box is what this work measures — one fixed view, perfect hindsight, or a choice a model had to learn — each judged on **both** success and cost.

